

Gaurav Ghati

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EDUCATION

University of California, Irvine

Sep 2023 – Dec 2024

Master of Computer Science, GPA: 3.8/4

Irvine, CA

Courses: Big Data, Transaction Processing & Distributed Data Management, Operating Systems, Data Structures

Pune Institute of Computer Technology

May 2018 – Jun 2022

Bachelor of Engineering in Information Technology, GPA: 9.21/10

Pune, India

TECHNICAL SKILLS

Skills: C++, Java, JavaScript, Python, Apache Spark, Kafka, Hadoop, Hive, Spring Boot, MySQL, MongoDB, Linux, Docker

AWS Services: RDS, DynamoDB, S3, EC2, Lambda, ECS, CloudFormation, Cloud Watch, Virtual Private Cloud

Domain: Big Data Management, Databases, Data Analytics, Machine Learning, Web Development, Microservices

EXPERIENCE

Amazon Web Services, Redshift Team

Feb 2025 – Present

Software Development Engineer

East Palo Alto, CA

- Led a security project to migrate from multi-tenant shared Fargate **compilation workers** to isolated single-invocation **Lambda** executions for the Redshift Compilation Service to eliminate possibility of cross-customer malicious attacks
- Designed a security validation layer for Redshift Compilation Service which compiles 14 million queries/day across 40+ AWS regions. Build with **Lambda warmpool architecture** and auto-scaler logic to handle 15k queries/min at peak workload
- Implemented a **patching mechanism** to patch 250+ ECS fargate instances by building **CICD pipeline** to continuously deploy patched images and a Lambda to update ECS services whenever there is a new image available
- Built a scalable, serverless **Redshift Internal Usage Dashboard** using AWS Lambda, DynamoDB, and CloudFormation to detect idle clusters across all regions - reducing operational costs by \$650K/month

Bank of New York Mellon

Jul 2022 – Aug 2023

Software Engineer

Pune, India

- Implemented backend microservices in Java, leveraging **Spring Boot** and **MySQL**, to enhance data delivery and quality for a **Data Warehouse**, now utilized by 10,000+ platform feeds
- Executed a migration from **MapReduce** to **Spark** within big data feeds, achieving a 45% runtime reduction and phasing out outdated technologies for streamlined operations
- Developed **Java** based **microservices** for data processing, integrating **Spark**, **HIVE**, and **SQL**, to manage over 100,000 records per dataset, incorporating design patterns and exception handling for improved data management

Veritas Technologies LLC

Aug 2021 – May 2022

Research Project Intern | [Github](#)

Pune, India

- Devised **NLP** text classification model using **LSTM** and Google's LLM - **BERT** for auto-matching documents. Utilized rehearsal techniques to prevent memory loss while training model
- Improved incremental learning recall by **45%** by writing a ranking algorithm prioritizing uncertain documents.
- Built web app with **Django** REST APIs in Python and **React.js**, the backend was **dockerized** and deployed to **AWS EC2** for scalable and efficient access

PROJECTS

TripTonic | Mobile App | [Github](#)

Jan 2024 - May 2024

- Led a team of four to build an **LLM-powered** mobile app that turns conversations into personalized trip plans, with a two-click generation flow in **React Native** and **Expo** requiring no setup
- Built recommendation engine using **GPT-3.5**, **LangChain**, **Google Maps** that adapts to budget, transport, and cuisines

Dynamic Thread Scheduler | *Operating Systems, Linux* | [Github](#)

Oct 2023 - Nov 2023

- Implemented thread scheduling in **C**, including thread structure creation and dynamic stack memory allocation
- Created **Round Robin thread scheduling** using a Circular Linked List and achieved concurrent non-parallel execution by context switching between threads with `setjmp()` and `longjmp()` **kernel** system calls

LEADERSHIP AND CO-CURRICULAR ACTIVITIES

- Served as a technical head of PICT IEEE Student Branch (PISB) for more than three years.
- Written technical blogs on medium.com/@gghati on topics like Natural Language Processing(NLP) models